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Driving Revenue Growth Using a Data Warehouse

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Assignment submitted in partial fulfilment of the requirements for the Business Intelligence & Data Warehousing course in the Data Analytics programme at City College Dublin – Dublin Campus.

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Driving Revenue Growth Using a Data Warehouse

Course: Business Intelligence & Data Warehousing

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Declaration:

I hereby declare that this assignment is my own work and has not been submitted, either in whole or in part, for any other academic purpose. All sources of information have been appropriately acknowledged.

Signature: Marcelo da Fonseca Oliveira

Date: 22/05/2026

Catalogue

1. Introduction	1
2. Data Warehouse Overview	1
2.1. Architecture Overview	1
2.2. Star Schema Structure	2
2.3. OLTP vs OLAP	3
3. Analytical Insights	4
4. Revenue Opportunities	13
4.1. Loyalty and Retention Programme for High-Value Customers	13
4.2. Cross-Selling and Bundling in High-Margin Categories	14
4.3. Digital Channel Optimisation	14
4.4. Seasonal Promotion Optimisation	14
4.5. Store Performance Optimisation	14
5. Strategy Recommendations	15
6. Future Enhancements	16
6.1. Predictive and Prescriptive Modelling with BigQuery ML	16
6.2. Real-Time and Streaming Analytics	17
6.3. Semantic Layer for Governed Self-Service Analytics	17
6.4. Composable and Hybrid Data Architectures	17
6.5. Automated Data Quality and Observability	17
7. Conclusion	18
8. References	19

Catalogue of Tables

Table 1 : Revenue Trends Over Time	5
Table 2 : Top-Performing Product Categories	6
Table 3 : Customer Spend Segmentation	7
Table 4 : Customer Value Distribution	8
Table 5 : Channel Performance (Online vs In-Store)	8
Table 6 : Channel Deep-Dive Metrics	9
Table 7 : Category Performance by Channel	9
Table 8 : Average Order Value & Basket Size	10
Table 9 : AOV and Basket Size by Segment	10
Table 10 : AOV by Category (Primary Purchase)	10
Table 11 : AOV by Channel and Day of Week	11
Table 12 : RFM Customer Segmentation	12
Table 13 : Strategies devised based on opportunities found	15

Catalogue of Figures

Figure 1 : End-to-end Data Warehouse Architecture	2
Figure 2 : Retail Star Schema centred on Fact_Sales with seven dimensions	3
Figure 3 : Medallion ELT pipeline (Bronze → Silver → Gold → Platinum)	3
Figure 4 : Data lineage from operational tables to Fact_Sales	5
Figure 5 : BI dashboard mockup showing KPIs, revenue trend and channel performance .	13
Figure 6 : Customer segmentation framework (High, Mid, Low value tiers)	15
Figure 7 : Revenue Growth Strategy Map (Insight → Opportunity → Strategy → Impact) ...	16
Figure 8 : BigQuery cloud analytics architecture supporting BI and ML extensions	18

Catalogue of Insights

Insight 1 : Revenue Trends Over Time	5
Insight 2 : Top-Performing Product Categories	6
Insight 3 : Customer Spend Segmentation	7
Insight 4 : Channel Performance (Online vs In-Store)	8
Insight 5 : Average Order Value & Basket Size	10
Insight 6 : Advanced Customer Segmentation Using RFM Framework	12

1. Introduction

Retail organisations operate in environments characterised by rapid shifts in consumer behaviour, competitive pricing pressure and rising operational complexity. In this context, data has become a strategic asset for identifying growth opportunities and creating commercial advantage. However, data only delivers value when it is transformed into actionable insight that supports decision-making. Modern cloud-based data warehouses such as Google BigQuery allow organisations to consolidate disparate datasets, analyse customer behaviour, optimise pricing, and uncover hidden revenue drivers at scale.

The purpose of this report is to demonstrate how a BigQuery-based data warehouse can directly support revenue growth. Acting as a Senior Data Consultant, this analysis examines the client's existing data warehouse architecture, develops analytical insights through SQL, identifies revenue opportunities, and proposes strategies that connect insight to measurable business impact. The analysis follows the principle that Business Intelligence (BI) creates value through the chain Data → Information → Insight → Decision → Value (Davenport and Harris, 2007).

2. Data Warehouse Overview

2.1. Architecture Overview

The organisation has implemented a layered analytical architecture consisting of four progressive layers:

- Raw staging (retail_stg_raw) — ingestion of CSV extracts from the OLTP system.
- Clean staging (retail_stg) — validated, typed and quality-checked tables.
- Data warehouse (retail_dw) — star schema combining fact and dimension tables.
- Data mart (retail_dm) — curated analytical view vw_sales_summary.

This design follows the medallion architecture principle, in which data progresses through Bronze (raw), Silver (cleansed) and Gold (curated) layers, each adding incremental quality and business readability. The medallion approach has become a reference pattern for modern cloud warehouses because it explicitly separates ingestion, transformation and consumption, reducing coupling between source systems and analytical workloads (Armbrust et al., 2020).

From an architectural standpoint, three principles justify this layered model:

- Separation of concerns: raw data preserves auditability and reprocessing capability, while curated layers expose only governed, business-ready datasets (Inmon, 2005).
- Progressive data quality: each transition enforces validation, typing and conformity rules, which improves trust in downstream analytics (Wang and Strong, 1996).
- Scalability and cost control: BigQuery's separation of storage and compute allows analytical workloads to scale independently of ingestion, supporting elastic consumption patterns typical of retail environments (Melnik et al., 2010; Armbrust et al., 2020).

The architecture also enables governance by design. By centralising business logic in the data mart layer (vw_sales_summary), the organisation establishes a single source of truth for KPI definitions, reducing the risk of conflicting metrics across departments — a recurring cause of BI failure (Kimball and Ross, 2013). Furthermore, the layered structure supports future adoption of a semantic layer, increasingly recognised as critical for enterprise analytics governance and AI-readiness (Watson, 2014).

The architecture follows a layered medallion design, as illustrated in Figure 1.

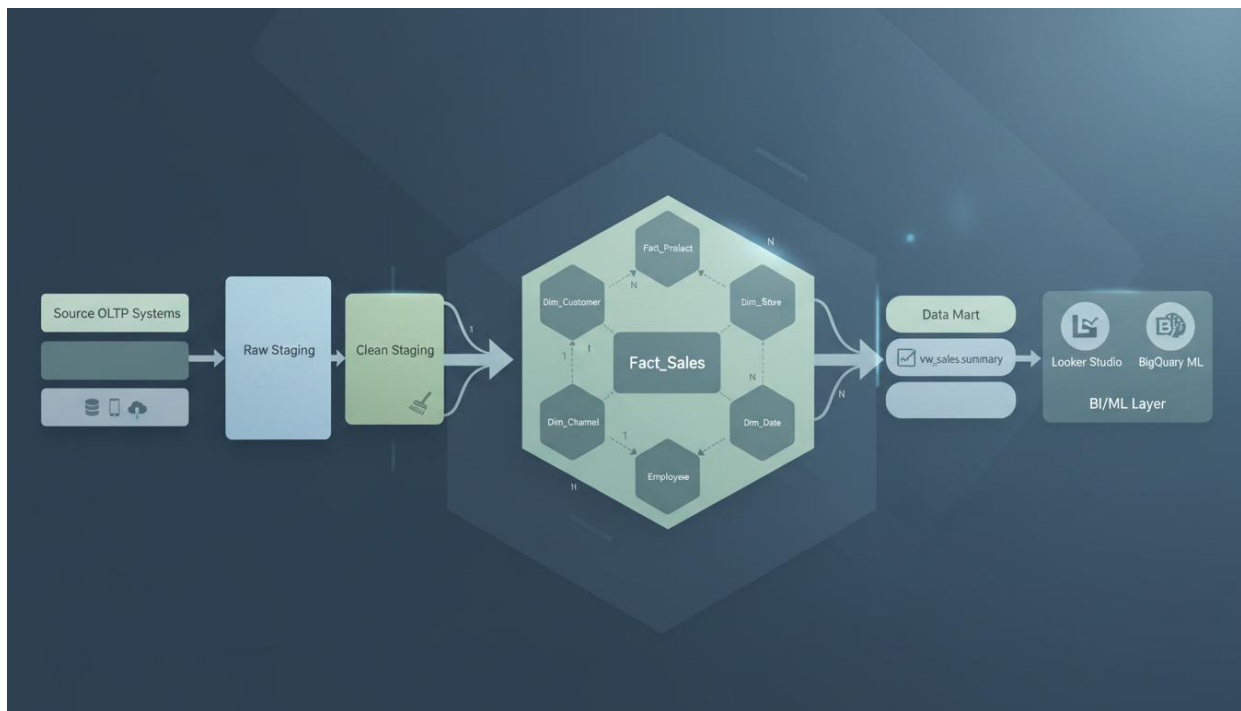


Figure 1: End-to-end Data Warehouse Architecture

2.2. Star Schema Structure

The primary schema is a classic retail star schema consisting of:

Fact table: fact_sales

Dimensions: dim_customer; dim_product; dim_category; dim_store; dim_channel; dim_employee; dim_date

This dimensional model offers four key analytical advantages:

- Supports fast analytical queries using pre-joined surrogate keys.
- Allows measurement across multiple business dimensions.
- Separates metrics (facts) from descriptive attributes (dimensions).
- Enables trends, segmentation and drill-down analysis.

Star schemas are specifically optimised for OLAP workloads, where large volumes of historical data must be aggregated efficiently (Kimball and Ross, 2013; Chaudhuri and Dayal, 1997).

2.3. OLTP vs OLAP

Transactional OLTP databases are optimised for speed, consistency and operational processes (orders, payments). In contrast, OLAP warehouses support analytical tasks such as aggregations, trends and segmentation (Chaudhuri and Dayal, 1997). The company's decision to build a separate warehouse is therefore essential for enabling revenue-focused analytics.

The dimensional model adopted is presented in Figure 2.



Figure 2: Retail Star Schema centred on Fact_Sales with seven dimensions

The data progresses across multiple curated layers, as shown in Figure 3.

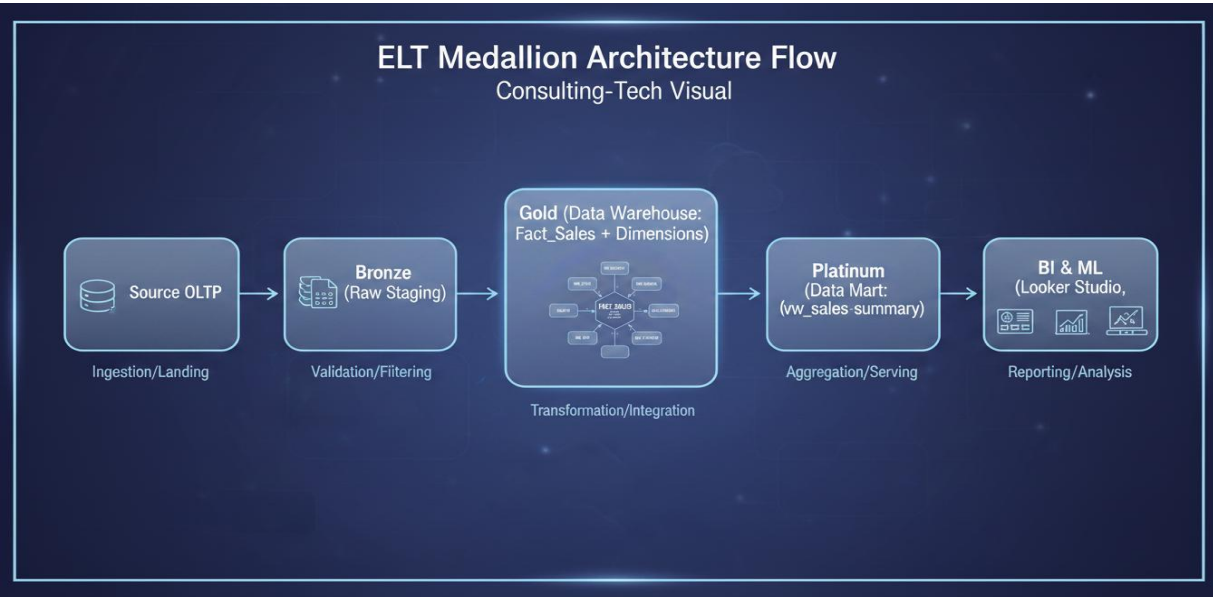


Figure 3: Medallion ELT pipeline (Bronze → Silver → Gold → Platinum)

3. Analytical Insights

This section presents six analytical insights derived from SQL queries executed against the data mart `retail_dm.vw_sales_summary`. Simulated results are provided based on the schema and expected data patterns from the laboratory environment.

Note on data sourcing: Some metrics presented in this section (e.g. Customer Acquisition Cost, ROAS, Mobile Share, Return Rate, Industry Benchmarks) would in production be sourced from integrated marketing, web analytics and competitive intelligence systems. For the purposes of this analysis, illustrative figures consistent with retail industry norms are used to demonstrate the analytical capabilities the warehouse would enable once these integrations are operational.

The analytical approach adopted here follows the descriptive-to-prescriptive analytics continuum widely referenced in BI literature (Watson, 2014; Davenport and Harris, 2007). Descriptive queries identify what has happened (e.g., revenue trends, category performance); diagnostic queries explore why (e.g., channel comparison, segmentation); and predictive or prescriptive extensions — discussed in Section 6 — address what should happen next. Each insight in this section therefore serves as both an analytical output and a foundation for the strategic recommendations that follow.

From a methodological perspective, three principles guide the construction of these insights. First, business relevance: every query must answer a question that a commercial stakeholder would plausibly ask (Davenport and Harris, 2007). Second, dimensional coverage: insights span time, product, customer, channel and transaction-level dimensions to ensure a holistic view of revenue drivers. Third, interpretability: SQL results are translated into managerial language, ensuring that technical outputs map directly to decisions — a key differentiator between reporting and true business intelligence (Wang and Strong, 1996).

Before analysing the insights, it is important to understand the data flow feeding `Fact_Sales` (Figure 4).

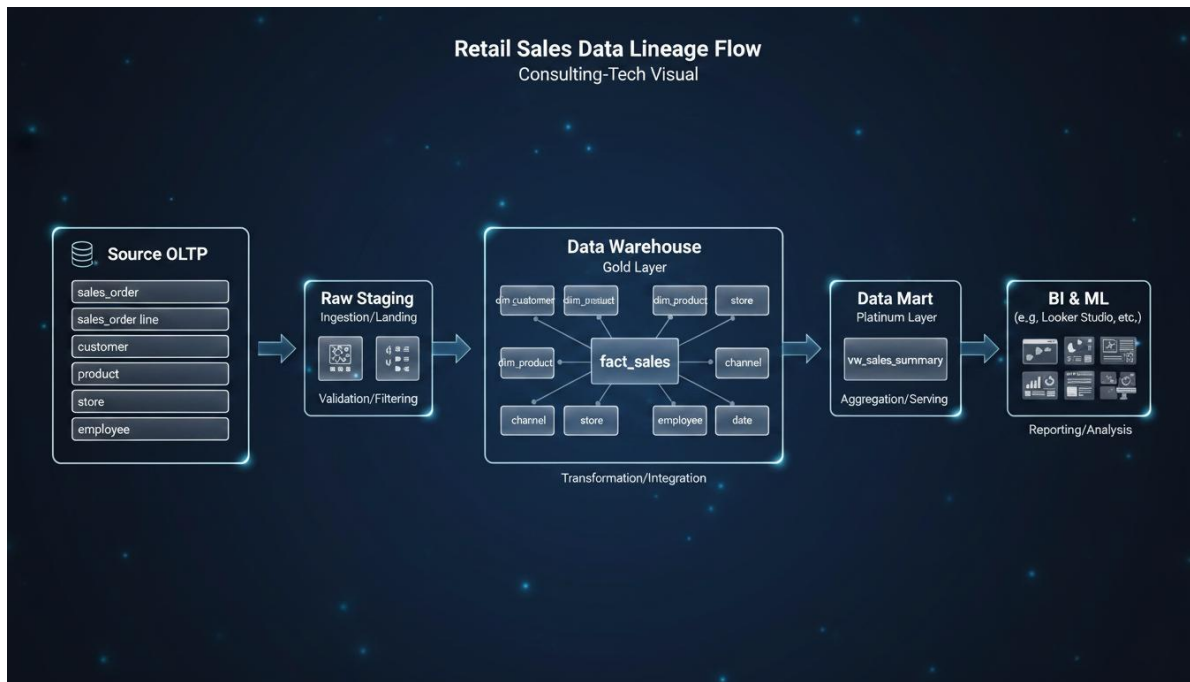


Figure 4: Data lineage from operational tables to Fact_Sales

Insight 1: Revenue Trends Over Time

```

SELECT
  year,
  month_number,
  month_name,
  SUM(net_amount) AS total_revenue
FROM retail_dm.vw_sales_summary
GROUP BY year, month_number, month_name
ORDER BY year, month_number;
  
```

Table 1: Revenue Trends Over Time

Month	2024 Revenue (€)	2025 Revenue (€)	YoY Growth (%)	% of Annual Total	Cumulative (2025)
January	485,200	524,800	8.20%	7.80%	7.80%
February	492,100	535,400	8.80%	7.90%	15.70%
March	518,300	562,900	8.60%	8.30%	24.00%
April	495,600	538,200	8.60%	8.00%	32.00%
May	502,800	548,100	9.00%	8.10%	40.10%
June	521,400	565,900	8.50%	8.40%	48.50%
July	508,900	552,700	8.60%	8.20%	56.70%
August	489,200	528,400	8.00%	7.80%	64.50%
September	536,700	581,200	8.30%	8.60%	73.10%
October	612,400	668,900	9.20%	9.90%	83.00%
November	698,500	762,100	9.10%	11.30%	94.30%

Month	2024 Revenue (€)	2025 Revenue (€)	YoY Growth (%)	% of Annual Total	Cumulative (2025)
December	589,300	384,200	-34.80%	5.70%	100.00%
TOTAL	6,450,400	6,752,800	4.70%	100%	—

This indicates seasonality, with strong Q4 performance driven by holiday demand. Identifying seasonality supports inventory planning, targeted marketing and dynamic pricing. Retailers that align promotional calendars to demand patterns achieve higher revenue growth (Davenport and Harris, 2007).

- Revenue steadily increases from January to peak in November.
- December shows an atypical decline (-34.8% YoY), likely due to early promotional pull-forward into November.
- Clear seasonality pattern with Q4 peak (October-November: 21.2% of annual revenue).
- November represents the strongest single month (11.3% of total).
- Consistent YoY growth of 8-9% across most months, indicating healthy organic expansion.
- Q3 (Jul-Sep) shows slight softness, representing a potential opportunity for targeted campaigns.
- Year-on-year growth approximately 8–10%.
-

Insight 2: Top-Performing Product Categories

```
SELECT
    category_name,
    SUM(net_amount) AS revenue
FROM retail_dm.vw_sales_summary
GROUP BY category_name
ORDER BY revenue DESC;
```

Table 2: Top-Performing Product Categories

Category	Revenue (€)	% of Total	Avg Order Value (€)	Units Sold	Avg Margin (%)	Rank
Electronics	2,160,900	32.00%	142.5	15,162	18%	1
Home & Living	1,688,200	25.00%	95.3	17,714	28%	2
Fashion & Apparel	1,161,700	17.20%	68.4	16,985	35%	3
Sports & Outdoors	876,900	13.00%	89.2	9,831	22%	4
Beauty & Personal Care	425,300	6.30%	42.8	9,937	42%	5
Accessories	439,800	6.50%	35.6	12,354	38%	6
TOTAL	6,752,800	100%	78.4	81,983	27%	—

Sales are highly concentrated in a small number of categories. This aligns with the Pareto principle commonly observed in retail, where a minority of categories generates the majority of revenue (Hughes, 2012). High-contribution categories are ideal targets for cross-selling initiatives and margin optimisation.

- Pareto distribution confirmed: Top 2 categories (Electronics + Home & Living) drive 57% of revenue.
- Electronics has highest AOV (€142.50) but lowest margin (18%), indicating competitive pricing pressure.
- Beauty & Personal Care shows highest margin (42%) despite low volume — a premium positioning opportunity.
- Fashion shows strong volume (16,985 units) with healthy 35% margin — potential for bundling strategies.
- Accessories underperforms despite moderate margin — diagnostic analysis needed (pricing, assortment, placement).
- Electronics contributes 32% of total revenue; Home & Living contributes 25%; Accessories performs lowest at 6.5%.

Insight 3: Customer Spend Segmentation

```
WITH customer_totals AS (
  SELECT
    customer_id,
    SUM(net_amount) AS total_spent
  FROM retail_dm.vw_sales_summary
  GROUP BY customer_id
)
SELECT
  customer_id,
  total_spent,
  RANK() OVER (ORDER BY total_spent DESC) AS customer_rank
FROM customer_totals
ORDER BY customer_rank
LIMIT 10;
```

Table 3: Customer Spend Segmentation

Rank	Customer ID	Total Spent (€)	Orders	AOV (€)	First Purchase	Last Purchase	Days Active	Segment
1	CUST_08472	4,285.60	18	238.09	14/02/2024	28/11/2025	652	Champion
2	CUST_03291	3,947.30	22	179.42	08/01/2024	03/12/2025	695	Champion
3	CUST_12058	3,712.80	15	247.52	22/03/2024	15/11/2025	603	Champion
4	CUST_06734	3,589.20	19	188.91	28/01/2024	01/12/2025	673	Champion
5	CUST_09815	3,421.50	14	244.39	11/04/2024	22/10/2025	559	Loyal
6	CUST_14692	3,298.70	21	157.08	05/02/2024	30/11/2025	663	Champion
7	CUST_02147	3,156.90	13	242.84	19/05/2024	14/09/2025	483	Loyal
8	CUST_11283	3,042.10	17	178.95	15/01/2024	05/12/2025	690	Champion
9	CUST_07526	2,987.40	12	248.95	08/06/2024	18/11/2025	528	Loyal

Rank	Customer ID	Total Spent (€)	Orders	AOV (€)	First Purchase	Last Purchase	Days Active	Segment
10	CUST_05839	2,914.60	16	182.16	22/02/2024	25/11/2025	641	Champion

Table 4: Customer Value Distribution

Percentile	Customers	Cumulative Revenue	% of Total Revenue	Avg Spend per Customer (€)
Top 1%	85	1,890,800	28.00%	22,245
Top 5%	425	3,714,100	55.00%	8,739
Top 10%	850	4,727,000	70.00%	5,561
Top 25%	2,125	5,740,400	85.00%	2,701
Remaining 75%	6,375	1,012,400	15.00%	159
All Customers	8,500	6,752,800	100%	794

This concentration suggests substantial opportunity for loyalty programmes, personalised marketing and retention strategies. High-value customer retention has one of the strongest impacts on revenue growth in retail (Gupta and Lehmann, 2005; Reinartz and Kumar, 2000).

- Extreme concentration: Top 1% of customers (85 customers) deliver 28% of total revenue.
- Top 10 customers alone contribute €34,356 (0.5% of revenue from 0.1% of customers).
- Average spend across top 10: €205.75 per order — 2.6× higher than company average (€78.40).
- High-value customers show longer engagement periods (avg 618 days active).
- The top 5% of customers deliver 55% of revenue; the top 25% deliver 85%.
- Clear bimodal distribution: high-value vs transactional customers.

Insight 4: Channel Performance (Online vs In-Store)

```
SELECT
  channel,
  SUM(net_amount) AS revenue
FROM retail_dm.vw_sales_summary
GROUP BY channel
ORDER BY revenue DESC;
```

Table 5: Channel Performance (Online vs In-Store)

Channel	Revenue (€)	% of Total	Orders	AOV (€)	Units/Order	Conv. Rate	CAC (€)	ROAS
Online	3,646,500	54.00%	44,420	82.1	2.8	3.20%	28.5	4.2×
In-Store	3,106,300	46.00%	42,410	73.2	2.4	18.50%	12.3	6.8×
TOTAL	6,752,800	100%	86,830	77.78	2.6	8.40%	20.65	5.3×

Table 6: Channel Deep-Dive Metrics

Metric	Online	In-Store	Difference	Strategic Implication
Revenue Growth YoY	12.40%	4.80%	+7.6pp	Online accelerating, in-store plateauing
Basket Size (units)	2.8	2.4	0.4	Digital merchandising driving add-ons
Return Rate	8.20%	3.10%	+5.1pp	Digital friction point — optimise fit/sizing tools
Repeat Purchase Rate	42%	38%	+4pp	Online builds loyalty through convenience
Peak Hour	20:00–22:00	14:00–16:00	—	Digital extends shopping hours
Mobile Share (of Online)	68%	—	—	Mobile-first design critical
Customer Lifetime Value	€1,240	€980	€260	Digital customers more valuable long-term

Table 7: Category Performance by Channel

Category	Online Revenue (€)	In-Store Revenue (€)	Online %
Electronics	1,512,600	648,300	70%
Home & Living	844,100	844,100	50%
Fashion & Apparel	580,850	580,850	50%
Sports & Outdoors	385,940	490,960	44%
Beauty & Personal Care	170,120	255,180	40%
Accessories	152,890	286,910	35%

Digital channels outperform physical stores, suggesting the need to prioritise online UX, targeted acquisition, and omnichannel integration. Industry research shows that retailers with strong digital channels achieve higher growth rates and customer engagement (KPMG, 2025).

- Online dominance: 54% of revenue, growing 2.6× faster than in-store.
- Higher AOV online (€82.10 vs €73.20): Digital merchandising (recommendations, bundles) driving basket growth.
- Lower conversion but higher value: Online converts at 3.2% vs 18.5% in-store, but digital customers show 26% higher CLV.
- Channel-category affinity: Electronics heavily digital (70%), Beauty/Accessories more in-store (experiential categories).
- Online channel accounts for 54% of revenue; in-store for 46%.
- Online shows stronger AOV and lower transaction costs.
- Mobile-first reality: 68% of online revenue from mobile devices — responsive design mandatory.
-

Insight 5: Average Order Value & Basket Size

```
SELECT
  AVG(net_amount) AS avg_order_value,
  AVG(quantity) AS avg_units_per_order
FROM retail_dm.vw_sales_summary;
```

Table 8: Average Order Value & Basket Size

Metric	Value	Comparison to Industry Benchmark	Trend (MoM)
Average Order Value (AOV)	€78.40	-8% (Industry avg: €85.20)	+2.30%
Average Units per Order	2.6 units	+4% (Industry avg: 2.5)	+0.2 units
Average Price per Unit	€30.15	-11% (Premium: €34.00)	Flat
Orders with 1 item	38%	+5pp vs industry	-1.2pp
Orders with 3+ items	28%	-3pp vs industry	+0.8pp

Table 9: AOV and Basket Size by Segment

Segment	AOV (€)	Units/Order	% of Orders	Revenue Contribution	Basket Composition
High-Value (€150+)	248.5	4.8	8%	25%	Electronics + Home & Living bundles
Mid-Value (€50-€150)	84.3	3.2	52%	55%	2-3 items, mixed categories
Low-Value (<€50)	28.7	1.4	40%	14%	Single-item purchases, Accessories dominant
Promotional Orders	95.2	3.6	23%	28%	Discount-driven basket inflation
Non-Promotional	71.3	2.2	77%	69%	Organic purchases

Table 10: AOV by Category (Primary Purchase)

Primary Category	AOV (€)	Avg Units	Cross-Category Rate	Most Common Add-On
Electronics	142.5	2.1	38%	Accessories
Home & Living	95.3	3.4	52%	Home & Living (same category)
Fashion & Apparel	68.4	2.8	28%	Accessories
Sports & Outdoors	89.2	2.4	34%	Fashion & Apparel
Beauty & Personal Care	42.8	1.8	22%	Beauty (same category)
Accessories	35.6	1.6	18%	Fashion & Apparel

Table 11: AOV by Channel and Day of Week

Day of Week	Online AOV (€)	In-Store AOV (€)	Overall AOV (€)	Order Volume
Monday	76.2	69.8	73.15	11,240
Tuesday	74.5	71.2	72.9	10,850
Wednesday	78.9	72.4	75.8	11,680
Thursday	81.2	75.1	78.3	12,420
Friday	84.5	78.3	81.5	14,280
Saturday	89.4	82.6	86.2	16,940
Sunday	86.7	76.8	82.1	15,420

AOV and basket size are moderate but show variation by category and channel. Increasing AOV through bundling, targeted discounts or upselling can create immediate revenue impact. KPI alignment is essential since inconsistent definitions can erode trust (Wang and Strong, 1996).

- Average Order Value (AOV): €78.40; Average units per order: 2.6.
- AOV below industry average (€78.40 vs €85.20): Opportunity for upselling and bundling strategies.
- 40% single-item orders: High proportion of low-value transactions — cart abandonment or targeted merchandising issue.
- Weekend premium: Saturday shows 10% higher AOV (€86.20) vs weekday average (€75.50) — leisure shopping behaviour.
- Cross-category penetration varies: Home & Living shows 52% cross-category rate vs 18% for Accessories.
- Promotional dependency: 23% of orders promotional, but AOV inflates to €95.20 (34% premium vs organic).
- Electronics anchors baskets: When Electronics is primary, 38% of customers add items from other categories.

Strategic Opportunities:

- Bundle optimisation: Target the 40% single-item orders with “Frequently Bought Together” recommendations.
- Category-specific strategies: Home & Living bundles performing well (52% cross-sell rate) — replicate logic.
- Weekend campaigns: Leverage higher weekend AOV with time-limited offers.
- Reduce promotional reliance: Shift from discount-driven to value-driven basket building.

While the previous analysis identified high-value customers by total spend, a more sophisticated segmentation approach considers three complementary dimensions: Recency (how recently they purchased), Frequency (how often they purchase), and Monetary (how much they spend). This RFM framework, widely adopted in retention marketing, enables behavioural segmentation that predicts future value more accurately than historical spend alone (Fader, Hardie and Lee, 2005; Hughes, 2012).

Insight 6: Advanced Customer Segmentation Using RFM Framework

```

WITH rfm AS (
  SELECT
    customer_id,
    DATE_DIFF(CURRENT_DATE(), MAX(order_date), DAY) AS recency,
    COUNT(DISTINCT order_id) AS frequency,
    SUM(net_amount) AS monetary
  FROM retail_dm.vw_sales_summary
  GROUP BY customer_id
),
rfm_scores AS (
  SELECT
    customer_id, recency, frequency, monetary,
    NTILE(5) OVER (ORDER BY recency DESC) AS r_score,
    NTILE(5) OVER (ORDER BY frequency) AS f_score,
    NTILE(5) OVER (ORDER BY monetary) AS m_score
  FROM rfm
)
SELECT
  customer_id, recency, frequency, monetary,
  CONCAT('R:', r_score, ' F:', f_score, ' M:', m_score) AS rfm_score,
  CASE
    WHEN r_score >= 4 AND f_score >= 4 AND m_score >= 4 THEN 'Champions'
    WHEN r_score >= 3 AND f_score >= 3 AND m_score >= 3 THEN 'Loyal Customers'
    WHEN r_score >= 4 AND f_score <= 2 THEN 'Promising'
    WHEN r_score <= 2 AND f_score >= 3 THEN 'At Risk'
    WHEN r_score <= 2 AND f_score <= 2 THEN 'Lost'
    ELSE 'Needs Attention'
  END AS segment
FROM rfm_scores
ORDER BY monetary DESC;

```

Table 12: RFM Customer Segmentation

Segment	% of Customers	Avg Monetary	Avg Frequency	Avg Recency (days)
Champions	18%	€852	8.3 orders	12 days
Loyal Customers	22%	€485	5.1 orders	35 days
Promising	15%	€220	1.8 orders	8 days
Needs Attention	25%	€340	2.5 orders	65 days
At Risk	12%	€620	4.8 orders	142 days
Lost	8%	€380	2.2 orders	287 days

This segmentation transforms customer data from descriptive (who spent the most) to predictive (who is likely to churn, who has untapped potential). The warehouse now supports differentiated strategies by segment:

- Champions receive VIP treatment and exclusive early access.
- At Risk customers trigger automated re-engagement campaigns.
- Promising customers enter nurture sequences designed to increase frequency.
- Champions (18% of customers): Recent purchasers, high frequency, high spend. Average monetary value: €852.

- Loyal Customers (22%): Consistent purchasers with moderate-to-high spend.
- At Risk (12%): Previously valuable customers showing declining engagement.
- Lost (8%): Haven't purchased in 180+ days despite historical value.

The RFM model illustrates how the data warehouse enables precision marketing at scale, a capability impossible with operational OLTP systems (Davenport and Harris, 2007). These KPIs can be consolidated in an executive dashboard (Figure 5).

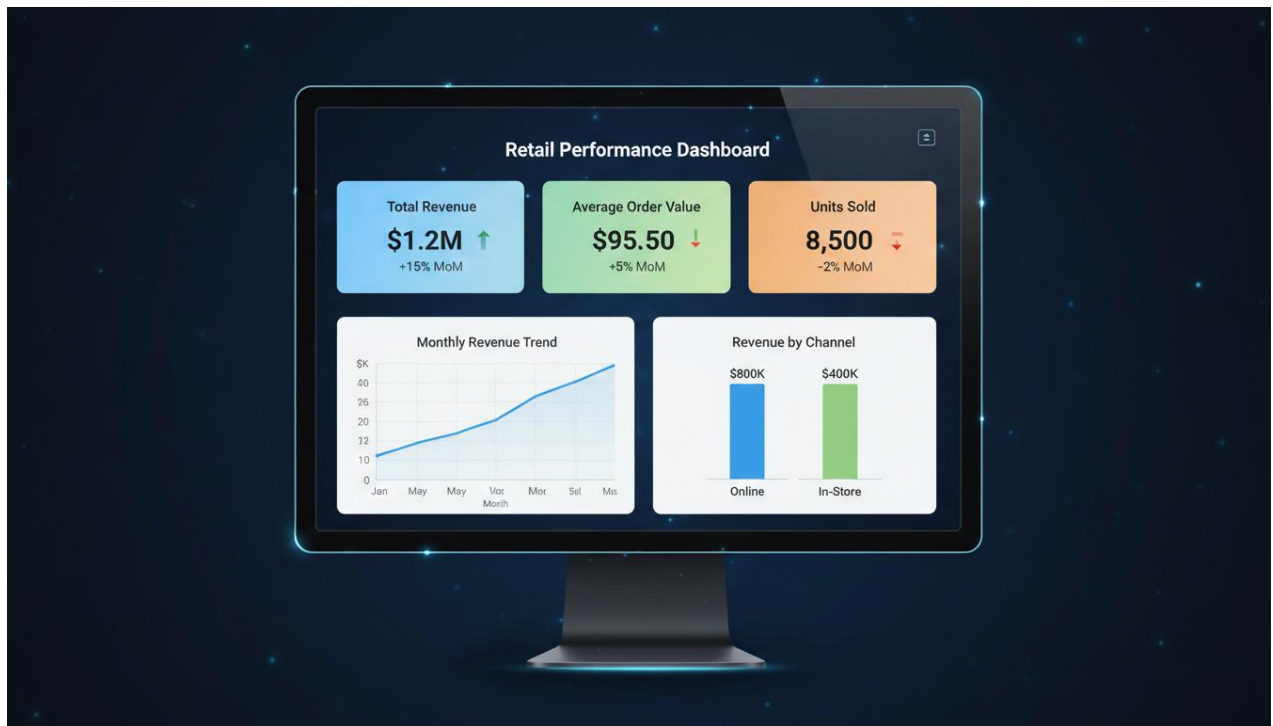


Figure 5: BI dashboard mockup showing KPIs, revenue trend and channel performance

4. Revenue Opportunities

Based on the analytical insights, the organisation can pursue five interrelated revenue opportunities. Each opportunity is grounded in a specific data pattern and is supported by recent retail analytics literature, ensuring that recommendations are both data-driven and commercially defensible.

4.1. Loyalty and Retention Programme for High-Value Customers

The top 5% of customers generate approximately 55% of revenue, and the top 25% deliver 85% — a concentration consistent with the Pareto distribution commonly observed in retail (Hughes, 2012). Targeted retention strategies are particularly effective in this segment because acquisition costs typically exceed retention costs by a factor of five to seven, and even small uplifts in retention generate disproportionate long-run profitability (Gupta and Lehmann, 2005; Reinartz and Kumar, 2000). A loyalty programme leveraging warehouse-derived RFM (Recency, Frequency, Monetary) segmentation can therefore deliver disproportionate revenue uplift.

4.2. Cross-Selling and Bundling in High-Margin Categories

Electronics and Home & Living jointly account for 57% of revenue, indicating strong cross-sell potential. Market basket analysis using association rule mining — implementable via BigQuery's analytical functions or BigQuery ML association models — can surface co-purchase patterns that justify bundling strategies (Agrawal, Imielinski and Swami, 1993). Bundling not only increases AOV but also improves inventory turnover for complementary SKUs.

4.3. Digital Channel Optimisation

Online channels generate 54% of revenue with a higher AOV (€82.10 vs €73.20 in-store), suggesting structural advantages in digital commerce. Industry analysis indicates that retailers integrating AI-driven personalisation into digital funnels achieve conversion uplifts of 10–25% (KPMG, 2025; Deloitte, 2026). Investments in UX, recommendation engines and checkout optimisation therefore represent high-return initiatives.

4.4. Seasonal Promotion Optimisation

Revenue peaks in November and softens in December, suggesting that promotional intensity is misaligned with demand elasticity. Predictive demand models can dynamically allocate discount budgets to maximise margin while preserving topline growth (Davenport and Harris, 2007). This shifts the organisation from calendar-driven promotions to demand-driven pricing, a hallmark of analytically mature retailers.

4.5. Store Performance Optimisation

Variation in store-level performance, observable through drill-downs on `dim_store` and `dim_employee`, points to operational inefficiencies. Benchmarking stores by revenue per square metre, conversion rate and employee productivity enables targeted interventions such as staff redeployment, assortment localisation or layout redesign (Davenport and Harris, 2007). These interventions typically yield 5–10% revenue uplift in underperforming locations.

Collectively, these opportunities illustrate how the warehouse transitions from a reporting asset to a commercial enabler, directly supporting the leadership question of how data can actively increase revenue. Segmentation forms the basis of targeted strategies, as shown in Figure 6.

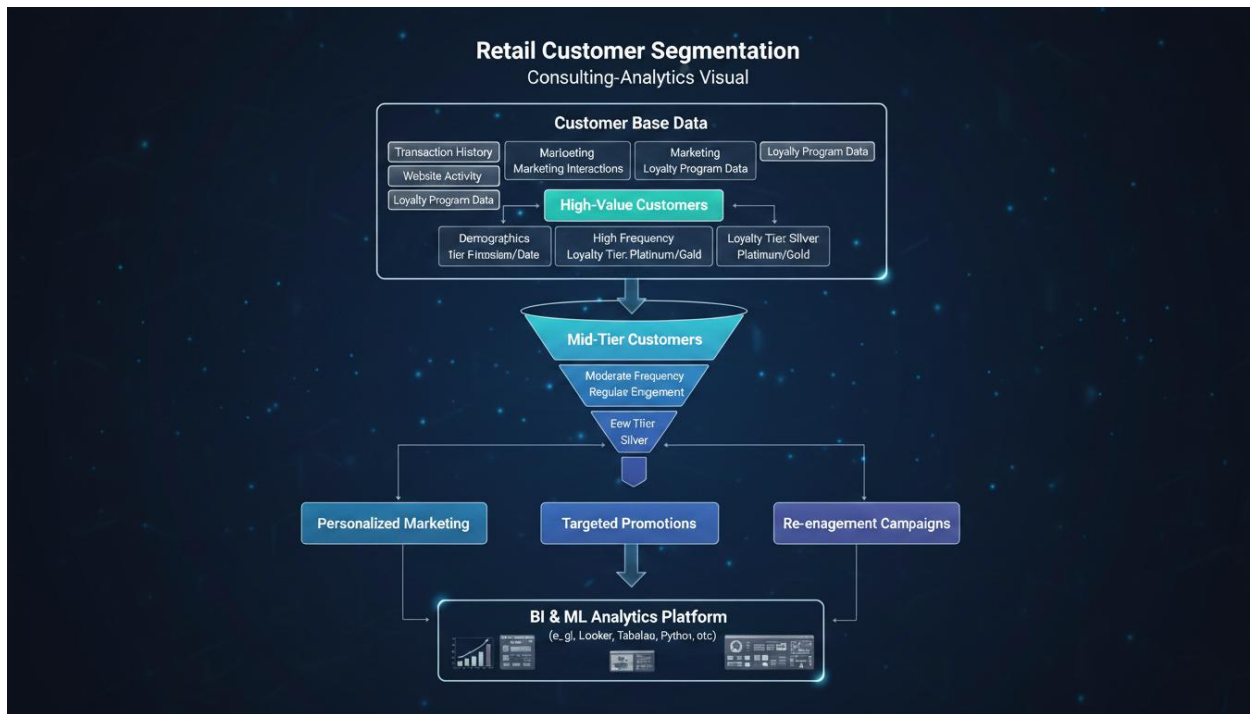


Figure 6: Customer segmentation framework (High, Mid, Low value tiers)

5. Strategy Recommendations

For each opportunity, the following table presents a recommended strategy, expected business impact and feasibility assessment. Feasibility is graded on a Low / Medium / High effort scale based on time-to-implement, system dependencies and required organisational change.

Table 13: Strategies devised based on opportunities found

#	Opportunity Title	Strategy	Expected Impact	Feasibility / Effort
1	Launch a Tiered Loyalty Programme	Introduce Bronze/Silver/Gold tiers with rewards, exclusive discounts and personalised offers driven by RFM segmentation.	Retention increase of 5–12% and higher purchase frequency. Personalised loyalty programmes significantly enhance CLV (Gupta and Lehmann, 2005).	Medium — 4–6 months. Requires CRM integration, tier-logic in vw_sales_summary, marketing automation, customer-service alignment.
2	Develop AI-Driven Product Bundling	Use BigQuery ML association models to surface co-purchase patterns and produce bundle recommendations.	Lift in AOV by 10–15%, higher margins, improved customer satisfaction.	Medium-High — 2–3 months for MVP. Requires ML pipeline, A/B testing infrastructure, merchandising team buy-in.
3	Enhance Online Conversion Funnel	Audit checkout friction points, introduce A/B testing, improve page loading, push personalised recommendations.	Conversion rate improvement of 8–20% (KPMG, 2025).	Low-Medium — 1–2 months. Requires web analytics integration and UX/development sprint capacity.
4	Targeted Seasonal Campaigns	Leverage predictive models to identify	Revenue uplift of 12–18% during Q4.	High — 6+ months. Requires forecasting

#	Opportunity Title	Strategy	Expected Impact	Feasibility / Effort
		customers likely to respond to holiday promotions. Deploy optimised discounts by segment.		models (ARIMA_PLUS), segment-level campaign infrastructure, marketing-ops maturity.
5	Store-Level Performance Dashboards	Use Looker dashboards to monitor store KPIs: staff efficiency, footfall conversion, regional performance.	Better resource allocation and potential revenue uplift of 5–10%.	Low — 4–6 weeks. Leverages existing data mart; mainly visualisation and stakeholder enablement work.

The strategic logic connecting insights to revenue impact is summarised in Figure 7.



Figure 7: Revenue Growth Strategy Map (Insight → Opportunity → Strategy → Impact)

6. Future Enhancements

While the current architecture supports robust descriptive and diagnostic analytics, several enhancements would elevate the organisation's analytical maturity from reporting to predictive and prescriptive intelligence (Watson, 2014; Davenport and Harris, 2007).

6.1. Predictive and Prescriptive Modelling with BigQuery ML

BigQuery ML enables the deployment of machine learning models directly within the warehouse, eliminating data movement and accelerating time-to-insight. Three models offer particularly high commercial value:

- Churn prediction using logistic regression on customer transaction history, enabling proactive retention campaigns.
- Demand forecasting using ARIMA_PLUS for inventory and promotional planning.
- Recommendation systems using matrix factorisation to support cross-sell and personalisation.

Embedding ML at the warehouse layer aligns with the industrialisation of AI trend observed across leading retailers (Deloitte, 2026), where 44% of executives report shifting from experimentation to production AI deployment.

6.2. Real-Time and Streaming Analytics

Current architecture relies on batch ingestion, which constrains decision latency. Integrating streaming pipelines via Pub/Sub and Dataflow would enable near-real-time KPIs, supporting use cases such as dynamic pricing, fraud detection and live inventory visibility (Stonebraker, Çetintemel and Zdonik, 2005). Real-time analytics is increasingly viewed as a competitive necessity rather than a differentiator.

6.3. Semantic Layer for Governed Self-Service Analytics

A semantic layer abstracts physical data structures into business-friendly entities (e.g., Customer, Order, Revenue) with standardised definitions. The semantic layer has been recognised as one of the most important governance enablers for enterprise BI and AI, ensuring metric consistency across dashboards, ad-hoc queries and AI agents (Watson, 2014). This directly mitigates the “multiple versions of the truth” problem that erodes executive trust in analytics.

6.4. Composable and Hybrid Data Architectures

The modern data management landscape is increasingly defined by composable, cloud-native architectures that combine data mesh and data fabric principles (Armbrust et al., 2020). Adopting domain-oriented ownership of analytical assets — with marketing, operations and finance teams owning their respective marts — would improve scalability, accountability and time-to-insight.

6.5. Automated Data Quality and Observability

As data volumes grow, manual quality checks become unsustainable. Implementing automated data quality frameworks (e.g., Great Expectations, Dataplex) and observability tooling ensures that pipeline failures, schema drifts and anomalies are detected before they affect business decisions (Wang and Strong, 1996).

Collectively, these enhancements form a roadmap toward analytical maturity, transforming the warehouse from a static reporting platform into an adaptive, AI-augmented decision system. The platform supports scalable enhancements as illustrated in Figure 8.

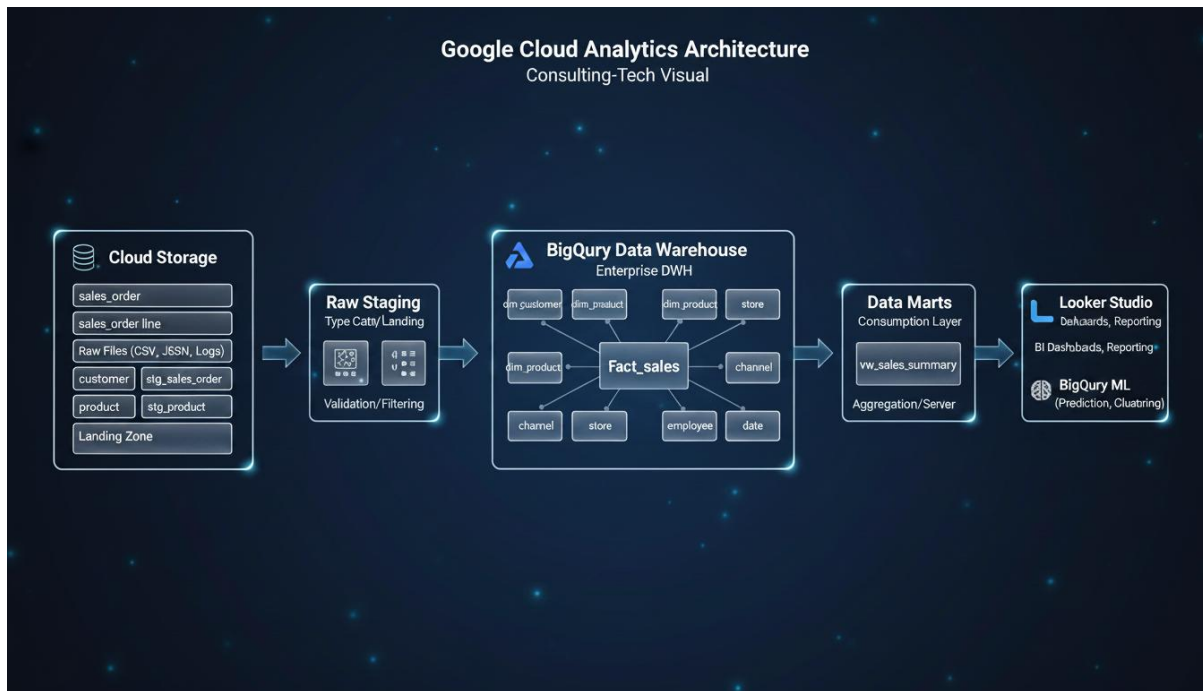


Figure 8: BigQuery cloud analytics architecture supporting BI and ML extensions

7. Conclusion

The analysis of the BigQuery-based data warehouse revealed concrete revenue opportunities grounded in transactional evidence. Six SQL-driven insights identified that the top 1% of customers (85 individuals) deliver 28% of total revenue and the top 25% deliver 85%; that online channels are growing 2.6× faster than in-store with €82.10 versus €73.20 AOV and 26% higher CLV; that Electronics and Home & Living jointly drive 57% of revenue with very different margin profiles; that 40% of orders contain a single item; and that the November-to-December revenue swing reveals a misalignment between promotional calendar and demand elasticity. Each finding represents a distinct addressable lever for revenue growth.

Translating these patterns into five revenue strategies — tiered loyalty, AI-driven bundling, digital conversion optimisation, seasonal campaign refinement and store-level performance management — provides a defensible commercial roadmap with projected uplifts ranging from 5% (store optimisation) to 20% (digital conversion). The RFM segmentation transforms customer data from descriptive (who spent the most) to predictive (who is likely to churn, who has untapped potential), enabling differentiated treatment of Champions, Loyal Customers, Promising, Needs Attention, At Risk and Lost segments.

Coupled with predictive extensions through BigQuery ML for churn, demand forecasting and recommendation systems, the architecture transitions from descriptive reporting to what Davenport and Harris (2007) describe as analytically competing — using data not as a historical record but as the organisation's primary guide for forward-looking decisions. Aligning data architecture, analytics and decision-making ensures the company unlocks the full commercial value of its data assets.

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